

The Feynman path integral

I) Background on Gaussian integrals

Prop: Let $A \in GL_n(\mathbb{C})$ be symmetric and pos. definite. Then $\forall y \in \mathbb{R}^n$,

$$\int_{\mathbb{R}^n} e^{-\frac{Ax \cdot x}{2}} e^{iy \cdot x} d^n x = \frac{(2\pi)^{\frac{n}{2}}}{\sqrt{\det A}} e^{-\frac{A^{-1}y \cdot y}{2}}.$$

If $A=1, n=1$ then $f(y) := \text{LHS}$ satisfies

$$f'(y) = y f(y), \quad f(0) = \sqrt{2\pi},$$

so $f(y) = \sqrt{2\pi} e^{-\frac{y^2}{2}}.$

For other A, n , let C be a pos. square root of A . Let $v = Cx$. Then

$$\text{LHS} = \int e^{-\frac{1}{2}|v|^2} e^{iC^{-1}y \cdot v} \frac{dv}{\det C} \quad \leftarrow \sqrt{\det A}$$

$\uparrow$ $C^2 x = Cx \cdot Cx$ as $C^T = C$

A is diagonalisable (it defines an inner product) with pos. eigenvalues

This is a product of 1D-integrals which equals the RHS.

Note: By analytic continuation, this also holds for

$$\{A \in GL_n(\mathbb{C}) : A^T = A, \operatorname{Re} A \text{ pos. def.}\}.$$

Also, by taking $A = A' + \varepsilon I$, the result holds for $\operatorname{Re} A'$ pos. semidefinite, where both sides are distributions.

Cor: With the same assumptions,

$$\int_{\mathbb{C}^n} e^{-Az \cdot z} e^{i(z \cdot \bar{w} + \bar{z} \cdot w)} d^n z = \frac{\pi^n}{\det A} e^{-A^{-1}w \cdot \bar{w}}.$$

$\swarrow$ volume on $\mathbb{C}^n$

Prop: With the same assumptions,

$$\int x_{i_1} \dots x_{i_{2k}} e^{-\frac{Ax \cdot x}{2}} d^n x = \frac{(2\pi)^{\frac{n}{2}}}{\sqrt{\det A}} \sum \alpha_{i_1 i_2} \alpha_{i_3 i_4} \dots \alpha_{i_{2k-1} i_{2k}},$$

$\swarrow$ over all unordered pairs of i 's

where $A^{-1} = (\alpha_{lm}).$ $\uparrow$ the integrand vanishes if this is odd

$$\begin{aligned} \text{LHS} &= (-1)^k \frac{\partial^{2k}}{\partial y_{i_1} \dots \partial y_{i_{2k}}} \bigg|_{y=0} \int e^{-\frac{Ax \cdot x}{2}} e^{iy \cdot x} d^n x \\ &= (-1)^k \frac{(2\pi)^{\frac{n}{2}}}{\sqrt{\det A}} \frac{\partial^{2k}}{\partial y_{i_1} \dots \partial y_{i_{2k}}} \bigg|_{y=0} e^{-\frac{A^{-1}y \cdot y}{2}}, \text{ and analyse...} \end{aligned}$$

II) Background on functional derivatives

Let $\mathcal{X}$ be a space of functions on $\mathbb{R}^n$, $f \in \mathcal{X}$, $\Phi: \mathcal{X} \rightarrow \mathbb{C}$. We define

$$\frac{\partial \Phi}{\partial f}: \mathcal{X} \rightarrow \mathbb{C}, \quad g \mapsto \lim_{\varepsilon \rightarrow 0} \frac{\Phi(f + \varepsilon g) - \Phi(f)}{\varepsilon},$$

whenever it exists.

Examples:

$$\cdot \tilde{\Phi}: f \mapsto \int f(y) h(y) dy$$

$$\leadsto \frac{\delta \tilde{\Phi}}{\delta f}(g) = \lim \int g h = \tilde{\Phi}(g)$$

$$\cdot \Phi: f \mapsto \exp \tilde{\Phi}(f)$$

$$\leadsto \frac{\delta \Phi}{\delta f}(g) = \tilde{\Phi}(g) \Phi(f) = \tilde{\Phi}(g) \exp \tilde{\Phi}(f)$$

$$\leadsto \frac{\delta^3 \Phi}{\delta f^3}(\delta_{x_1}, \dots, \delta_{x_3}) = h(x_1) \dots h(x_3) \exp \int h f$$

$$\cdot \tilde{\tilde{\Phi}}: f \mapsto \int f(y) K(y, z) f(z) dy dz \quad \text{with } K(x, y) = K(y, x)$$

$$\begin{aligned} \leadsto \frac{\delta \tilde{\tilde{\Phi}}}{\delta f}(g) &= \lim \frac{1}{\varepsilon} \int K(y, z) [(f(y) + \varepsilon g(y))(f(z) + \varepsilon g(z)) - f(y)f(z)] \\ &= \lim \int K(y, z) [g(y)f(z) + g(z)f(y)] \\ &= 2 \int K(y, z) f(y) g(z) \quad \text{since } K(y, z) = K(z, y) \\ &= 2 \int \underbrace{\left(\int K(y, z) f(y) dy \right)}_{h(z)} g(z) dz \end{aligned}$$

$$\cdot \Phi: f \mapsto \exp \tilde{\tilde{\Phi}}(f)$$

$$\leadsto \frac{\delta \Phi}{\delta f}(g) = 2 \int K(y, z) f(y) g(z) \Phi(f) = \frac{\delta \tilde{\tilde{\Phi}}}{\delta f}(g) \Phi(f)$$

III) A quantum-mechanical particle in $\mathbb{R}^2$

We now study the solutions to the Schrödinger equation $\partial_t \Psi = -i\hbar H \Psi$, where

$$H = -\frac{1}{2m} \nabla^2 + V(x).$$

To do this, we will compute the integral kernel for e^{-itH} . First, some notation:

Let $|x\rangle = \delta(x-\cdot)$, $|p\rangle = e^{ip\cdot}$. Then

$$\int |x\rangle \langle x| dx : f \mapsto \int |x\rangle \langle x| f \rangle = \int f(x) \delta(x-\cdot) = f,$$

$$\int |p\rangle \langle p| \frac{\partial p}{(2\pi)^2} : f \mapsto \frac{1}{(2\pi)^2} \int |p\rangle \langle p| f \rangle = \frac{1}{(2\pi)^2} \int \hat{f}(p) e^{ip\cdot} = f.$$

Let $|x,t\rangle = e^{itH} |x\rangle$. Then

$$\Psi(t,x) = \langle x| e^{-itH} |\Psi\rangle = \langle x,t | \Psi \rangle,$$

$$\int |x,t\rangle \langle x,t| dx : f \mapsto f \quad \forall t.$$

Let $t_f > t_i$. Then

$$\Psi(x, t_f) = \langle x, t_f | \Psi \rangle = \int \langle x, t_f | y, t_i \rangle \langle y, t_i | \Psi \rangle dy = \int \langle x, t_f | y, t_i \rangle \Psi(y, t_i) dy,$$

ie $\langle x, t_f | y, t_i \rangle$ is the integral kernel for $e^{-i(t_f-t_i)H}$. To compute it, let $\tau = \frac{t_f-t_i}{N}$, and $t_i = t_0 < t_0 + \tau < \dots < t_0 + N\tau = t_f$. Expand

$$\langle x_f, t_f | x_i, t_i \rangle = \int \dots \int \langle x_f, t_f | x_{N-1}, t_{N-1} \rangle \dots \langle x_1, t_1 | x_i, t_i \rangle dx_1 \dots dx_{N-1}.$$

Then, expand $e^{-i\tau H} = 1 - i\tau H + o(\tau)$ to calculate

$$\langle x_{n+1}, t_{n+1} | x_n, t_n \rangle = \langle x_{n+1} | e^{-i\tau H} | x_n \rangle$$

$$\begin{aligned} & \int \langle x_{n+1} | p \rangle \langle p | \nabla^2 | p \rangle \langle p | x_n \rangle \frac{\partial p \partial \bar{p}}{(2\pi)^2} \\ &= \int e^{ip(x_{n+1}-x_n)} |p|^2 \frac{\partial p}{(2\pi)^2} \end{aligned}$$

$$\begin{aligned} &= \delta(x_{n+1}-x_n) - i\tau \left(-\frac{1}{2m} \langle x_{n+1} | \nabla^2 | x_n \rangle + \langle x_{n+1} | V | x_n \rangle \right) + o(\tau) \\ &= \int e^{-ip(x_{n+1}-x_n)} \frac{\partial p}{(2\pi)^2} \left(1 - i\tau \left(\frac{|p|^2}{2m} + V(x_n) \right) \right) + o(\tau) \end{aligned}$$

$$\begin{aligned} x &= p, y = -(x_{n+1}-x_n) \\ A &= \frac{i\tau}{m} \end{aligned}$$

$$= e^{-i\tau V(x_n)} \int e^{-i\tau \frac{|p|^2}{2m}} e^{-ip(x_{n+1}-x_n)} \frac{\partial p}{(2\pi)^2} + o(\tau)$$

Plugging this into the formula for the Gaussian integral, with $x=p$, $y=-(x_{n+1}-x_n)$, and $A = \frac{i\hbar}{m}$, yields

$$\langle x_{n+1} t_{n+1} | x_n t_n \rangle = \left(\frac{m}{2\pi i \hbar} \right)^{\frac{1}{2}} \exp i \hbar \left[\frac{m}{2} \left| \frac{x_{n+1} - x_n}{\hbar} \right|^2 - V(x_n) \right] + o(\hbar).$$

or another constant?

Hence,

$$\langle x_f t_f | x_i t_i \rangle = \left(\frac{m}{2\pi i \hbar} \right)^{\frac{N}{2}} \int \dots \int \exp i \hbar \left[\sum_{n=0}^{N-1} \left(\frac{m}{2} \left| \frac{x_{n+1} - x_n}{\hbar} \right|^2 - V(x_n) \right) \right] dx_1 \dots dx_N + o(1)$$

Letting $\hbar \rightarrow 0$, we formally get

$$\langle x_f t_f | x_i t_i \rangle = \text{const.} \int \exp \left(i \int_{t_i}^{t_f} \left(\frac{m}{2} |\dot{x}(t)|^2 - V(x(t)) \right) dt \right) [dx],$$

an integral over the space of all paths, by interpreting $x_n = x(t_n)$. Note that the measure $[dx]$ does not really exist, and neither does the constant.

Note: Using the path integral for $\langle y + \Delta y, t + \Delta t | y, t \rangle$, one can rederive Schrödinger's equation.

IV) The generating functional

The path integral allows us to incorporate a lot of information into one functional.

Let $Z[F] = \int e^{i \int (L(x, \dot{x}) + F(t) \cdot x) dt} [dx],$

for $F: \mathbb{R} \rightarrow \mathbb{R}^2$. Then

$$\frac{\delta^3}{\delta F_{i_1} \dots \delta F_{i_3}} (t_1, \dots, t_3) Z = i^3 \int x_{i_1}(t_1) \dots x_{i_3}(t_3) Z[F] [dx].$$

S's at those points

Lemma: Let $X = (X_1, \dots, X_2)$ be the position operators, $X_t = e^{itH} X e^{-itH}$. Let $g_j: \mathbb{R}^2 \rightarrow \mathbb{R}$. Then $\langle x_f t_f | \underbrace{\prod g(X_{t_j})}_{G(X)} | x_i t_i \rangle = \underbrace{\int \prod g_j(x_{t_j})}_{G(x)} e^{i \int L(x, \dot{x}) dt} [dx]$. *paths from x_i to x_j*

We calculate

$$\begin{aligned} \langle x_f t_f | g(X_{t_i}) | x_i t_i \rangle &= \int \langle x_f t_f | g(X_{t_i}) | x_{t_i} t_i \rangle \langle x_{t_i} t_i | x_i t_i \rangle dx_{t_i} \\ &= \int g(x_{t_i}) \underbrace{\langle x_f t_f | x_{t_i} t_i \rangle}_{\int e^{i \int_{t_i}^{t_f} L(x, \dot{x}) dt} [dx]} \langle x_{t_i} t_i | x_i t_i \rangle dx_{t_i} \\ &= \text{RHS, since we also integrate over } x_{t_i}. \end{aligned}$$

The general case is similar.

Cor: For arbitrary states Ψ_i, Ψ_f , *over all paths*

$$\langle \Psi_f | \underbrace{\prod G(X)}_{G(X)} | \Psi_i \rangle = \int \Psi_f(x(t_f, t_f))^* \Psi_i(x(t_i, t_i)) G(x) e^{i \int L(x, \dot{x}) dt} [dx].$$

$\int \langle \Psi_f | x_f t_f \rangle \langle x_f t_f | \prod G(X) | x_i t_i \rangle \langle x_i t_i | \Psi_i \rangle dx_i dx_f$

Thm: Assume H has a discrete spectrum with eigenvectors $|\Psi_n\rangle$, $H\Psi_0 = 0$. Then

$$\underbrace{G(X)}_{G(X)} \langle \Psi_0 | \prod X_{i_1, t_1} \dots X_{i_3, t_3} | \Psi_0 \rangle = (-i)^3 \frac{\delta^3}{\delta F_{i_1} \dots \delta F_{i_3}} (t_1, \dots, t_3) \frac{Z[F]}{Z[0]}.$$

For $T_- < t_i < t_f < T_+$, we have

$$\begin{aligned} \langle x_+ T_+ | \prod G(X) | x_- T_- \rangle &= \iint \langle x_+ T_+ | x_f t_f \rangle \langle x_f t_f | \prod G(X) | x_i t_i \rangle \langle x_i t_i | x_- T_- \rangle dx_i dx_f \\ &= \sum_{mn} \iint \Psi_m(x_i) \Psi_n(x_i) \Psi_m(x_f)^* \Psi_n(x_-)^* e^{i(t_f E_m - T_+ E_m - t_i E_n + T_- E_n)} \langle \dots \rangle \end{aligned}$$

vanishes when $n \neq 0$ and $T_\pm \rightarrow \pm \infty (t \rightarrow \pm \infty)$

Hence, $\lim_{T_\pm \rightarrow \pm \infty (t \rightarrow \pm \infty)} \frac{\langle x_+ T_+ | \prod G(X) | x_- T_- \rangle}{\Psi_0(x_+) \Psi_0(x_-)^*} = \langle \Psi_0, t_f | \prod G(X) | \Psi_0, t_i \rangle = \text{LHS as } |\Psi_0, t_i\rangle = |\Psi_0\rangle.$

Now plug in the formulas from the lemma and divide by this result for $G=1$ ($\langle \Psi_0 | \Psi_0 \rangle = 1$).

Note: F is called a **source**. It can be interpreted as providing the source and sink for particles entering and leaving an interaction.

Note: We give all Lagrangians a small imaginary part, so the $T_{\pm} \rightarrow \pm \infty (1 - i\epsilon)$ is incorporated in the integral, even though we only write $\int L(x, x') dt$.

The version of this result for fields, the **Gell-Mann-Low formula**, states that

$$\langle \Omega | T \phi_1(x_1) \dots \phi_n(x_n) | \Omega \rangle = \lim_{\epsilon \rightarrow 0} \frac{\langle 0 | T \phi_1(x_1) \dots \phi_n(x_n) \exp(-i \int_{-T}^T H_I(t) dt) | 0 \rangle}{\langle 0 | T \phi_1(x_1) \dots \phi_n(x_n) | 0 \rangle}$$

$\uparrow$ vacuum $H\Omega=0$ $\uparrow$ free vacuum $H_0|0\rangle=0$

Numerator and denominator can be calculated using Feynman diagrams with external lines corresponding to propagators (instead of creation/annihilation). The diagrams for the denominator have no external lines - they are **vacuum bubbles**.

Let $W(p_1, \dots, p_k) = \int \dots \int \langle \Omega | T \phi_1(x_1) \dots \phi_n(x_n) | \Omega \rangle e^{i \sum p_j^\mu x_j^\mu} dx_1 \dots dx_n$. One can show, using the Gell-Mann-Low formula, that

$$\begin{aligned} & \langle 0 | a_1(p_1^{\text{out}}) \dots a_J(p_J^{\text{out}}) S a_1^{\dagger}(p_1^{\text{in}}) \dots a_k^{\dagger}(p_k^{\text{in}}) | 0 \rangle \\ &= i^{J+k} \prod_{j=1}^J \frac{u_j^*(p_j^{\text{out}})}{\hat{\Delta}_j(p_j^{\text{out}})} \prod_{k=1}^k \frac{u_k^*(p_k^{\text{in}})}{\hat{\Delta}_k(p_k^{\text{in}})} W(p_1^{\text{out}}, \dots, p_J^{\text{out}}, -p_1^{\text{in}}, \dots, -p_k^{\text{in}}). \end{aligned}$$

This is the LSZ formula. It allows to calculate S -matrix elements from field vacuum expectation values.

Va) Field theory - a real scalar field

For a free real scalar field $\phi(t, x)$, the Lagrangian density is

$$\mathcal{L}(\phi, \partial\phi) = \frac{1}{2}((\partial\phi)^2 - m^2\phi^2).$$

We form the generating functional

$$Z[F] = \int \exp(i \int [\underbrace{\frac{1}{2}((\partial\phi)^2 - m^2\phi^2)}_{\substack{\text{classical field} \\ \text{vanishing at } \infty}} + \underbrace{F\phi}_{\substack{\text{chain} \\ \text{rule} \\ -\phi\partial^2\phi}}] d^4x) [d\phi],$$

Small imaginary part
 $\leadsto (m^2 - i\epsilon)\phi$

ie

$$\frac{Z[F]}{Z[0]} = \frac{\int \exp(i \int \frac{1}{2}(\phi(\partial^2 + m^2 - i\epsilon)\phi + F\phi) d^4x) [d\phi]}{\int \exp(i \int \frac{1}{2}(\phi(\partial^2 + m^2 - i\epsilon)\phi) d^4x) [d\phi]}.$$

By an ∞ -dim. version of the Gaussian integrals, this equals

$$\exp\left(\frac{i}{2} \int F (\partial^2 + m^2 - i\epsilon)^{-1} F d^4x\right),$$

a finite dim. integral! One can show that $(\partial^2 + m^2 - i\epsilon)^{-1}$ is convolution with the Feynman propagator Δ_F , so this equals

$$\exp\left(\frac{i}{2} \int F(x) \Delta_F(x-y) F(y) d^4x d^4y\right),$$

which can easily be functionally differentiated.

Vb) Field theory - a Yang-Mills field for U(1)

Here, the action is $S(A) = -\frac{1}{4} \int F_{\mu\nu} F^{\mu\nu} d^4x = \frac{1}{2} \int A_\mu (\partial^\mu \partial^2 - \partial^\mu \partial^\nu \partial_\nu) A_\mu d^4x$, so

$$Z[\mathcal{J}] = \int \exp i (S(A) + \int \mathcal{J}^\mu A_\mu d^4x) [dA],$$

where the source $\mathcal{J}$ is vector valued. Up to constants, this should equal

$$\exp \left(\frac{i}{2} \int \mathcal{J}^\mu(x) \Delta_{\mu\nu}(x-y) \mathcal{J}^\nu(y) d^4x d^4y \right)$$

fundamental solution for $-\partial^\mu \partial^2 + \partial^\mu \partial^\nu \partial_\nu$
→ does not exist!

The problem is that the space of all A is too big, and the integrand is closed under the orbits of $A_\mu \mapsto A_\mu + \partial_\mu \chi$.

Analogy: Let $\sigma: \mathbb{R} \rightarrow \text{Isom}(\mathbb{R}^n)$ with unbounded orbits. Assume $F: \mathbb{R}^n \rightarrow \mathbb{R}$ is invariant under σ . Let $M = h^{-1}(0)$ be a hypersurface intersecting each orbit once. Then

$$1 = \int_{\mathbb{R}} \delta(h) du = \int \delta(h(\sigma_{t(x)}(x))) \Delta(x) dt,$$

where $\Delta(x) = \left| \frac{d}{dt} \right|_{t=t(x)} h(\sigma_t(x))$ and $\sigma_{t(x)}(x) \in h^{-1}(0)$. Then

$$\int F(x) d^n x = \iint F(x) \delta(h(\sigma_{t(x)}(x))) \Delta(x) dt d^n x$$

invariant under σ

$$\stackrel{x=\sigma_{t(y)}(y)}{=} \iint F(y) \delta(h(y)) \Delta(y) dt d^n y = \int dt \int_{h^{-1}(0)} F(y) \Delta(y) dy.$$

volume of symmetry group
= ∞

independent of h
and possibly
finite

In our situation, we have an additive symmetry group $G = C^1(\mathbb{R}^4)/\mathbb{R}$. We fix a gauge by requiring that $\partial^\mu A_\mu - w = 0$, for a fixed $w \in C(\mathbb{R}^4)$. This corresponds to $h^{-1}(0)$, where $h(A) = \partial^\mu A_\mu - w$. Then

$$1 = \int_G \delta(h) [dA] = \int_G \delta(h(A^\chi)) \det \frac{\delta h(A^\chi)}{\delta \chi} [d\chi].$$

Faddeev-Popov determinant

$$= \partial^2, \text{ since } h(A^\chi) = \partial^\mu (A_\mu + \partial_\mu \chi) - w = h(A) + \partial^2 \chi$$

Inserting this into $Z[\mathcal{J}]$, we get

$$Z[\mathcal{J}] = (\det \partial^2) \iint \exp i (S(A) + \int \mathcal{J}^\mu A_\mu d^4x) \delta(h(A^\chi)) [dA] [d\chi]$$

$$\stackrel{A \mapsto A^\chi}{=} (\det \partial^2) \int [d\chi] \int \exp i (S(A) + \int \mathcal{J}^\mu A_\mu d^4x) \delta(\partial^\mu A_\mu - w) [dA].$$

invariant under G
also invariant if
 $\partial_\mu \mathcal{J}^\mu = 0$

Conceptually:
 A is only defined up to gauge $A_\mu \mapsto A_\mu + \partial_\mu \chi$; for $\int \mathcal{J}^\mu A_\mu d^4x$ to be well defined, we need that $\int \mathcal{J}^\mu \partial_\mu \chi d^4x = 0 \forall \chi \in C^1(\mathbb{R}^4)$. Using the chain rule, this means $\partial_\mu \mathcal{J}^\mu = 0$ (charge conservation)

Until now, $w \in \mathcal{C}(\mathbb{R}^4)$ was arbitrary, and this also works for weighted averages of such w . Taking the weights to be $e^{-i \int \frac{w^2}{2a} \partial^4 x}$, for $a \in \mathbb{R}^x$, we get

$$Z[J] = N(a) (\det \partial^2) \int [d\lambda] \int e^{i(S(A) + \int J^\mu A_\mu)} e^{-i \int \frac{w^2}{2a} S(\partial^\mu A_\mu - w)} [dw] [dA]$$

$$= \int e^{i(S(A) - \frac{1}{2a} \int (\partial^\mu A_\mu)^2 + \int J^\mu A_\mu)} [dA]$$

integrating over w , i.e. setting $w = \partial^\mu A_\mu$

$$= \int e^{i \int -\frac{1}{2} A_\mu (-g^{\mu\nu} \partial^2 + (1 - \frac{1}{2}) \partial^\mu \partial^\nu) A_\nu + J^\mu A_\mu} [dA]$$

this makes things possible

using the definition

$$S(A) = \frac{1}{2} \int A_\mu (g^{\mu\nu} \partial^2 - \partial^\mu \partial^\nu) A_\nu$$

Now, the operator has a fundamental solution with $\hat{\Delta}_{\mu\nu}^{(a)}(p) = \frac{1}{p^2 + i\epsilon} (g^{\mu\nu} - (1-a) \frac{p^\mu p^\nu}{p^2})$.

$$\text{Then } Z[J] = \int \exp\left(\frac{i}{2} \int J^\mu(x) \Delta_{\mu\nu}^{(a)}(x-y) J^\nu(y) \partial^4 x \partial^4 y\right).$$

Note that $a \in \mathbb{R}^x$ can still be chosen freely.

VII) Interlude: Integration of Grassmann-valued functions

To do fermionic functional integrals, we need to be able to integrate anti-commuting functions.

Let V be an $\mathbb{R}$ -vector space with basis $z_1, \dots, z_n$, $\mathcal{G} = \mathbb{C} \otimes \wedge V$. A general element $f \in \mathcal{G}$ has the form

$$f(z_1, \dots, z_n) = c + \sum_i c_i z_i + \sum_{i < j} c_{ij} z_i z_j + \dots + c_{1 \dots n} z_1 \dots z_n,$$

where the c 's are complex numbers. We define

$$\frac{\partial}{\partial z_j} z_1 \dots z_n = \begin{cases} 0 & \text{o/w} \\ (-1)^{l-1} z_1 \dots \hat{z}_l \dots z_n & \text{if } j = l. \end{cases}$$

Note that $\left\{ \frac{\partial}{\partial z_j}, z_k \right\} = \left(\frac{\partial}{\partial z_j} z_k + z_k \frac{\partial}{\partial z_j} \right) = \delta_{jk} \mathbb{1}$, $\left\{ \frac{\partial}{\partial z_j}, \frac{\partial}{\partial z_k} \right\} = \left\{ z_j, z_k \right\} = 0$.

We first define integration wrt z_1 . We require

integrating out a variable $\int \partial z_1 f(z_1, \dots, z_n) = g(z_2, \dots, z_n)$ for some g ,

translation $\int \partial z_1 (\alpha + (z_1 + \gamma)\beta) = \int \partial z_1 (\alpha + z_1 \beta)$ for γ not depending on z_1 .

Together with linearity wrt "+" and wrt right multiplication by $g(z_2, \dots, z_n)$, they imply

$$\left(\int \partial z_1 z_1 \right) = \left(\int \partial z_1 (z_1 + \gamma) \right) = \left(\int \partial z_1 z_1 \right) + \left(\int \partial z_1 \gamma \right),$$

ie $\left(\int \partial z_1 \right) \gamma = 0 \quad \forall \gamma(z_2, \dots, z_n)$. Hence, $\int \partial z_1 = 0$. Normalising $\int \partial z_1 z_1 = 1$, we get

$$\int \partial z_1 (\alpha + z_1 \beta) = \beta.$$

In other words, $\int \partial z_1 f(z_1, \dots, z_n) = \frac{df}{dz_1}$, integration is the same as differentiation. We now define

$$\int \dots \int \partial z_{i_1} \dots \partial z_{i_n} f(z_1, \dots, z_n) = \int \partial z_{i_1} \left(\dots \left(\int \partial z_{i_n} f(z_1, \dots, z_n) \right) \dots \right).$$

We see (from $\int \partial z_1 = \frac{d}{dz_1}$ e.g.) that changing the order of integration yields a minus sign, ie the ∂z_i should anticommute too. One can also check that

$$\int \dots \int \partial z_1 \dots \partial z_n f(z_1, \dots, z_n) = c_{1 \dots n}.$$

If $\eta_j = \sum M_{jk} z_k$ is another basis for V , we have $\eta_1 \dots \eta_n = \det M z_1 \dots z_n$. We want

$$\int \partial \eta_1 \dots \partial \eta_n \eta_1 \dots \eta_n = \int \partial z_1 \dots \partial z_n z_1 \dots z_n = 1, \text{ ie } \partial \eta_1 \dots \partial \eta_n = \det M^{-1} \partial z_1 \dots \partial z_n.$$

If V was a $\mathbb{C}$ -vector space then $\mathcal{G} = \mathbb{C} \otimes \wedge V$ has a basis $z_1, \dots, z_n, z_1^*, \dots, z_n^*$ of ± 1 -eigenvectors for the complexified complex structure. In this basis, the previous formulas are still valid.

Now let $A \in M_{n \times n}(\mathbb{C})$, $z^* A z = \sum z_j^* A_{jk} z_k$, $\iint [dz^*, dz] = \iint \dots \iint dz_n^* dz_n \dots dz_1^* dz_1$.

Prop: $\iint [dz^*, dz] e^{-z^* A z} = \det A$.

↑
this picks up the term of order $2n$

We only need to analyse the coefficient of the term of order $2n$ in $e^{-z^* A z}$.

Now let $V_{\mathbb{C}}$ have basis $\{z_i, z_i^*, \eta_i, \eta_i^*\}_{i=1}^n$ and write $\eta^* z = \sum \eta_j^* z_j$. Let A be symmetric and invertible.

Prop: $\iint [dz^*, dz] e^{-z^* A z + i(\eta^* z + z^* \eta)} = \det A e^{-\eta^* A^{-1} \eta}$.

Cor: $\iint [dz^*, dz] z_j^* z_k e^{z^* A z} = \det A \frac{\partial}{\partial \eta_j} \frac{\partial}{\partial \eta_k^*} e^{-\eta^* A^{-1} \eta} \Big|_{\eta_j = \eta_k^* = 0} = \det A (A^{-1})_{jk}$.

Finally, a Grassmann spinor is a 4-tuple $\eta = (\eta_1, \dots, \eta_4)^T$. Its Dirac adjoint is

$\bar{\eta} = \eta^\dagger \gamma^0 = (\eta_1^*, \dots, \eta_4^*) \gamma^0$ and the corresponding volume element is

$$\gamma^0 = \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix}$$

$$[d\bar{\eta} d\eta] = [d\eta^\dagger d\eta] = d\eta_1^* d\eta_1 \dots d\eta_4^* d\eta_4,$$

where we used that $\det \gamma^0 = 1$.

VII a) Fermionic functional integrals

In the previous section, the Grassmann numbers were indexed by $1, \dots, \dim V$. Now, we want to make sense of an integral of the form

$$Z[\eta] = \iint [\bar{\psi} \psi] \exp(i \int (\bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi + \bar{\eta}\psi + \bar{\psi}\eta) d^4x),$$

where ψ, η are Grassmann spinor valued and indexed by space-time. The value of this integral should be

$$i^\infty \det(i\gamma^\mu \partial_\mu - m) \exp(i \int \bar{\eta} (-i\gamma^\mu \partial_\mu + m)^{-1} \eta d^4x),$$

in analogy to its finite dimensional version. The (infinite) constant is $Z[0]$, so it will factor out. The operator $(-i\gamma^\mu \partial_\mu + m)^{-1}$ is given by convolution with the fundamental solution of $-i\gamma^\mu \partial_\mu + m$, i.e. the Dirac propagator $(i\gamma^\mu \partial_\mu + m)\Delta_F$. Functional differentiation yields, as in the boson case, vacuum expectations for products of fields, e.g. the corresponding quantum fields

$$\langle 0 | \psi(x) \bar{\psi}(y) | 0 \rangle = \int \iint [\bar{\psi} \psi] \psi(x) \bar{\psi}(y) e^{i \int \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi d^4x}.$$

This formalism now allows for interactions of bosons and fermions. For example, in QED we have

$$Z[J, \eta, \bar{\eta}]$$

$$= \iiint [\bar{\psi} \psi] [\bar{\psi} \psi] e^{i \int (-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi - e A_\mu \bar{\psi} \gamma^\mu \psi) d^4x} e^{i \int (J^\mu A_\mu + \bar{\eta}\psi + \bar{\psi}\eta) d^4x}.$$

To evaluate this, we can expand the interaction term $e^{-ie A_\mu \bar{\psi} \gamma^\mu \psi}$ and use the integrals that we know for the free fields.

We will now try to make mathematical sense of the huge Grassmannian appearing in the path integral.

VIIIb) Grassmann-valued functions

ie Grassmann numbers indexed by space-time

We first define a Grassmann algebra for countably many discrete generators.

Let's start with a separable Hilbert space $\mathcal{H}$ with a decomposition $\mathcal{H} = \mathcal{H}_+ \oplus \mathcal{H}_-$ and an antiunitary involution $v \mapsto v^*$ interchanging the factors. Let $\mathcal{H}_+$ have orthonormal basis ζ_n , so $\mathcal{H}_-$ has orthonormal basis ζ_n^* . We could form the (algebraic) exterior algebra $\Lambda_0 \mathcal{H}$, but it does not contain infinite sums like $\sum \zeta_n^* \zeta_n$ or exponentials, and neither does its completion (the Fermion Fock space over $\mathcal{H}$).

Instead, consider its (algebraic) dual $\Lambda' \mathcal{H} \cong \Lambda_0 \mathcal{H}$ via the inner product.

Then $\Lambda' \mathcal{H}$ is also an algebra: The (exterior) product of $\phi, \psi \in \Lambda' \mathcal{H}$ is the element $\chi \in \Lambda' \mathcal{H}$ s.t. $\chi|_M = \phi|_M \psi|_M$ for all finite dimensional $M \subset \mathcal{H}$.

The Berezin integral $\Lambda' \mathcal{H} \rightarrow \mathbb{C}$ is similarly defined: If $M_n = \text{Span}\{\zeta_1, \dots, \zeta_n, \zeta_1^*, \dots, \zeta_n^*\}$,

then

$$\iint [\zeta^* \zeta] \phi = \lim_{n \rightarrow \infty} \iint \dots \iint \zeta_n^* \zeta_n \dots \zeta_1^* \zeta_1 \phi|_{M_n},$$

whenever the limit exists.

Example: Let $\sum_n \zeta_n^* \zeta_n =: \zeta^* \zeta$. Then $\iint [\zeta^* \zeta] e^{\zeta^* \zeta} = 1$, since this is true for all its finite dim. versions.

For our $\mathcal{H}_+$ and $\mathcal{H}_-$, we will take $L^2(\mathbb{R}^4)$. We enlarge both spaces to $\tilde{\mathcal{H}}_{\pm}$ to contain the δ -functions (eg $\tilde{\mathcal{H}}_{\pm} = H^{-s}(\mathbb{R}^4)$, a Sobolev space for $s > 2$). We define the anti-unitary involution $\mathcal{H} \rightarrow \mathcal{H}$, $(f, g) \mapsto (g^*, f^*)$.

Let $\zeta(x), \zeta^*(x)$ denote $\delta(x - \cdot)$ in $\tilde{\mathcal{H}}_+, \tilde{\mathcal{H}}_-$, respectively. We will write

$$f = \int f(x) \zeta(x) dx, \quad g = \int g(x) \zeta^*(x) dx,$$

for $f \in \mathcal{H}_+, g \in \mathcal{H}_-$. We use the notation to express the exterior product of $f, g \in \mathcal{H}_+$ as

$$f g = \int f(x) \zeta(x) dx \int g(x) \zeta(x) dx = \iint f(x) g(y) \zeta(x) \zeta(y) dx dy = \iint \frac{f(x)g(y) - f(y)g(x)}{2} \zeta(x) \zeta(y) dx dy.$$

↑ exterior product

↑ exterior product

↑ pointwise product

It now remains to identify Grassmann integrals over space-time with Berezin integrals $\Lambda^1 \mathcal{H} \rightarrow \mathbb{C}$, i.e. we need to identify their integrands with elements of $\Lambda^1 \mathcal{H}$. Let's do this for degree 2:

The completion $\Lambda^2 \mathcal{H}_+$ of $\Lambda_0^2 \mathcal{H}_+$ consists of skew-symmetric functions $F \in L^2(\mathbb{R}^4 \times \mathbb{R}^4)$, i.e. bilinear integral kernels for operators T_F on $L^2(\mathbb{R}^4)$. For the example f, g above,

$$F = \frac{1}{2} (f(x)g(y) - f(y)g(x)), \quad T_F: h \mapsto \frac{1}{2} (\langle h|g \rangle f - \langle h|f \rangle g).$$

More generally, elements of $\Lambda^2 \mathcal{H}$ are $F \in \text{Mat}_{2 \times 2}(L^2(\mathbb{R}^4))$ with $F_{ij}(x,y) = -F_{ji}(y,x)$ corresponding to

$$\iint F_{11}(x,y) \zeta(x) \zeta(y) + F_{12}(x,y) \zeta(x) \zeta^*(y) + F_{21}(x,y) \zeta^*(x) \zeta(y) + F_{22}(x,y) \zeta^*(x) \zeta^*(y) dx dy,$$

and to

$$T_F: (h_1, h_2) \mapsto (T_{F_{11}} h_1 + T_{F_{12}} h_2, T_{F_{21}} h_1 + T_{F_{22}} h_2) \in L^2(\mathbb{R}^4) + L^2(\mathbb{R}^4).$$

In particular, for $(f_1, f_2), (g_1, g_2) \in \mathcal{H}_+ \oplus \mathcal{H}_-$, the exterior product corresponds to

$$F_{ij}(x,y) = \frac{1}{2} (f_i(x)g_j(y) - f_j(y)g_i(x)), \quad T_F(h_1, h_2) = (H_1, H_2),$$

$$\text{where } H_k = (\langle h_1|g_1 \rangle + \langle h_2|g_2 \rangle) f_k + (\langle h_1|f_1 \rangle + \langle h_2|f_2 \rangle) g_k.$$

Example: Let $F(x,y) = \frac{1}{2} \begin{pmatrix} 0 & -\delta(x-y) \\ \delta(x-y) & 0 \end{pmatrix}$. This comes from an exterior product of $(0, \zeta_n^*)$ and $(\zeta_n, 0)$. $\zeta_n^* \zeta_n = -\delta(x-y)$ etc

In particular,

$$\zeta^* \zeta = \sum_n \zeta_n^* \zeta_n = \iint \delta(x-y) \frac{\zeta^*(x)\zeta(y) - \zeta(x)\zeta^*(y)}{2} dx dy = \int \zeta^*(x) \zeta(x) dx,$$

$$\text{and} \quad \iint [\zeta \zeta \zeta^*] e^{-\int \zeta^*(x) \zeta(x) dx} = \iint [\zeta \zeta \zeta^*] e^{-\zeta^* \zeta},$$

which can formally be evaluated as above.

Other Grassmann space-time integrals can be interpreted in a similar way.